

Development of Digital Platform Decision-Making Planning in Water Resource Management for Local Administrative Organizations Chachoengsao Province, Thailand

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Abstract

Water User Organizations (WUOs) and Local Administrative Organizations (LAOs) in Thailand have legal responsibilities for local water resource management and for submitting water-related projects to the Thai Water Plan (TWP) system. However, many local organizations face limitations in baseline data, digital tools, and technical capacity required for effective water planning. This study developed “Waterplan RRU” a digital platform designed to support water resource planning and decision-making and implemented it in four subdistricts of Chachoengsao Province: Samet Tai and Samet Nuea in Bang Khla District, Plaeng Yao in Plaeng Yao District, and Khlong Khut in Ban Pho District. The study involved four LAOs and six registered WUOs and employed participatory action research and development evaluation. The research was conducted in three phases: stakeholder engagement and baseline assessment, platform design and validation, and platform implementation with community-based data verification. The baseline assessment identified 102 water sources with a combined potential capacity of 7,318,430 cubic meters, compared with total water demand of 83,241,498.95 cubic meters, indicating a water deficit of 75,923,068.95 cubic meters across the study areas. The mobile-first platform received a high level of architectural suitability from nine domain experts (Mean ($\bar{x}$) = 4.61, S.D. = 0.48). Training improved the water-management and platform-use competencies of community and LAO personnel (n = 65). End users (n = 94) rated the platform at a high level across functional, usability, performance, and security dimensions (Mean = 4.15–4.25). The results of the knowledge assessment of trainees before training and the training achievement assessment after training showed that post-training achievement increased. Specifically, trainees who received training on the platform for planning and decision-making in water resource management for water user organizations and local administrative organizations in Chachoengsao Province had a post-training mean score of 10.50 with a standard deviation of 1.58, compared to a pre-training mean score of 8.53 with a standard deviation of 3.08. This indicates a statistically significant difference at the .01 level between the pre-training scores and the post-training achievement scores using the digital platform for water management training in Chachoengsao Province.

Keywords: digital platform, water resource management, community-based research, water user organization

1. Introduction

1.1 Background and Rationale

Chachoengsao Province lies within the Bang Pakong River Basin in Eastern Thailand, an area that experiences recurring dry-season water shortages, rainy-season flooding, and saltwater intrusion from the Bang Pakong River into agricultural and household water supplies. These pressures are compounded by aging and undersized irrigation infrastructure, siltation of canals and reservoirs, and competition for water among agricultural, household, and industrial users. Communities in the province have also reported a lack of reliable spatial and quantitative data on local water sources, which has made it difficult for water user organizations (WUOs) and local administrative organizations (LAOs) to plan interventions or to justify budget requests to higher levels of government.

At the national level, the Water Resources Act B.E. 2561 (2018), Section 17, establishes a planning hierarchy in which locally identified water-management needs are expected to move from provincial sub-committees and river basin committees up to the National Water Resources Committee (ONWR) for approval and funding. The National Water Resources Master Plan (a 20-year plan spanning 2018–2037, revised in its first phase for 2023–2037 around five policy pillars) similarly assumes that local organizations can generate and submit credible project data through the Thai Water Plan (TWP) system operated by the Office of National Water Resources (NWRO). In practice, however, WUOs and LAOs in the study area lacked both the digital tools and the technical skills required to collect, organize, and submit this data, leaving locally identified needs largely absent from the formal planning pipeline despite the existence of a legal and administrative pathway for them.

1.2 Research Objectives

This study was designed to: (1) assess the water resource potential, water demand, and water balance of the study area; (2) develop local capacity among WUO members, community leaders, and LAO staff in water data collection, digital platform use, and water-management project planning; (3) design, develop, and evaluate a digital platform to support water resource planning and decision-making; and (4) evaluate the outcomes of platform and capacity-building implementation, including competency gains, user satisfaction, and the resulting water-management plans and TWP project submissions.

1.3 Research Questions

The study addressed three questions: What is the current water resource potential, demand, and balance in the study subdistricts, and what data and capacity gaps prevent WUOs and LAOs from acting on this information? What platform architecture and functions are required to support community-level water data collection and translation of that data into fundable project proposals? What are the measurable effects of platform and capacity-building implementation on user competency, satisfaction, and the flow of community-identified water projects into local development plans and the national TWP system?

1.4 Conceptual Framework

The study treats the water-planning gap as simultaneously a data problem, a capacity problem, and a governance-pathway problem. A digital platform alone was assumed to be insufficient; the conceptual framework therefore links platform development to parallel local capacity-building (in water data collection, platform use, and project writing) and to the existing formal pathway from community water-management plans, to LAO local development plans, to TWP project submissions. Sustainability was conceptualized from the outset around four dimensions – technology (platform hosting and maintenance), data (ownership and stewardship), personnel (local and LAO innovators), and network (coordination among NWRO, LAOs, and communities) – so that platform outcomes could be evaluated not only on technical performance but on their integration into this governance pathway.

1.5 Expected Benefits

The project was expected to produce a validated water resource baseline for the four subdistricts; a functioning digital platform for ongoing water data collection and planning; measurable gains in local technical capacity; a set of community-endorsed water management plans integrated into local development plans; and a body of experience on translating community-generated data into TWP-eligible project proposals that other LAOs in the region could draw on.

2. Literature Review

2.1 National Water Policy and the Thai Water Plan System

The Water Resources Act B.E. 2561 (2018) provides the legal basis for integrated water resources management in Thailand, establishing river basin committees and a National Water Resources Committee (ONWR) with authority over water allocation and planning. Section 17 of the Act sets out a project-approval chain running from provincial sub-committees through basin committees to national approval. The National Water Resources Master Plan (20-year, 2018–2037, with its first phase revised for 2023–2037 around five pillars) operationalizes this framework, and the Thai Water Plan (TWP) system, administered by the Office of National Water Resources (NWRO) at <https://onwr.thaiwaterplan.com>, is the digital mechanism through which local organizations submit water-management projects for budget consideration. Access to and effective use of the TWP system by LAOs is therefore a precondition for community water needs to reach formal planning and funding processes.

2.2 Study Area Profile

The study was conducted in four subdistricts of Chachoengsao Province: Samet Tai and Samet Nuea subdistricts in Bang Khla District, Plaeng Yao subdistrict in Plaeng Yao District, and Khlong Khut subdistrict in Ban Pho District. All four areas lie within or near the Bang Pakong River Basin and share exposure to seasonal water shortage, flooding, and saltwater intrusion, though each subdistrict presents a distinct combination of agricultural land use, canal infrastructure, and water source types.

2.3 Water Resource Potential in the Bang Pakong Basin

Regional hydrological studies of the Bang Pakong basin have modeled discharge as a power function of catchment area ($QF = 7.0678 \cdot A^{0.8414}$, $R^2 = 0.9306$), reflecting the basin's mixed rainfall-runoff and tidal characteristics. Such basin-level models provide useful context for interpreting local water balance findings but are not a substitute for subdistrict-level source inventories, which this study undertook directly.

2.4 Water User Organizations

Water user organizations (WUOs) in Thailand are formally established under the Water User Organization Ministerial Regulation B.E. 2564 (2021), which sets out registration, governance, and operating requirements. As of 2 February 2023, 3,507 WUOs were registered nationally. Registered status gives WUOs formal standing to participate in local water-management planning and, in principle, in project submission processes, but registration alone does not confer the data, digital skills, or platform access needed to act on that standing – the gap this study addresses.

2.5 Community-Based Research and Experiential Learning

The study's methodological orientation draws on participatory action research (PAR) as a cyclical process of joint problem definition, action, and reflection with community stakeholders, and on community-based research (CBR) more broadly as a framework in which the community is a co-producer of knowledge rather than a passive subject of study. Capacity-building activities were further informed by experiential and lifelong-learning principles (following Kolb's experiential learning cycle), which shaped the study's hybrid, action-oriented training design combining onsite instruction with on-demand video resources and hands-on data collection.

2.6 Digital Water Governance

The study situates its platform-development work within the broader literature on water governance and digital government. The OECD Principles on Water Governance emphasize legitimacy, transparency, and stakeholder engagement in water decision-making, while the integrated water resources management (IWRM) literature frames water planning as inherently multi-sectoral and multi-scalar, requiring coordination across administrative levels. Work on digital government and data governance highlights that digital tools are necessary but not sufficient for improved governance outcomes; they must be paired with institutional arrangements for data ownership, stewardship, and use. These themes connect directly to Sustainable Development Goal 6 (clean water and sanitation), which calls for integrated water resources management and stakeholder participation, and they inform this study's emphasis on pairing platform development with capacity-building and a defined governance pathway from community to LAO to national systems.

3. Method

3.1 Research Design

This study used a participatory action research (PAR) design combined with development evaluation (DE), framed operationally as community-based research (CBR) integrated with a research-and-development (R&D) process for the digital platform itself. The PAR component structured ongoing cycles of joint problem identification, action (data collection, training, platform testing), and reflection with WUO members, community leaders, and LAO staff. The DE component was used to track and adapt the platform-development process itself as it moved from prototype to field deployment.

3.2 Study Area and Partners

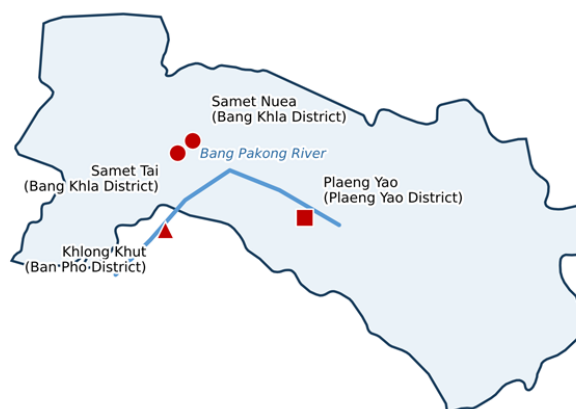
The study partnered with four LAOs under signed memoranda of understanding: Samet Tai Subdistrict Administrative Organization and Samet Nuea Subdistrict Administrative Organization (both Bang Khla District), Plaeng Yao Subdistrict Administrative Organization (Plaeng Yao District), and Khlong Khut Subdistrict Administrative Organization (Ban Pho District), all in Chachoengsao Province. Within these four subdistricts, six registered water user organizations participated directly (registration status as of 15 August 2024): Nong Sano

Phatthana 2 WUO (Moo 5, Samet Tai), Nong Sano WUO (Moo 2, Samet Tai), Kaset Ban Wang Si Thong WUO (Moo 6, Samet Tai), and Samet Tai WUO (Moo 6, Samet Tai) – all four in Samet Tai subdistrict, Bang Khla District; Ban Lat Bua WUO (Moo 4, Khlong Khut subdistrict, Ban Pho District); and Ban Phai Kao WUO (Moo 6, Plaeng Yao subdistrict, Plaeng Yao District). The location of the study area is presented in Figure 1.

(a) Location of Chachoengsao Province, Thailand



(b) Approximate locations of the four study subdistricts



Schematic diagram; not to scale and not for navigational use.

Figure 1. Study area map showing the location of Chachoengsao Province, Thailand, and the approximate locations of the four study subdistricts (Samet Tai, Samet Nuea, Plaeng Yao, and Khlong Khut)

3.3 Research Phases

The study was implemented in three phases. Phase 1 (Months 1–3) covered stakeholder consultation, the signing of MOUs between the university and the four LAOs, and baseline analysis of local water sources and sector-based water demand. Phase 2 (Months 4–6) covered digital-platform design and prototype development, training on the NWRO TWP system delivered by NWRO Region 2 (led by Regional Director Sarawut Sakol), platform prototype testing with LAO civil-engineering staff, and a nine-expert validation panel convened at the Chao Sawoei meeting room, Rajabhat Rajanagarindra University. Phase 3 (Months 7–12) covered platform rollout, community-level data collection and verification, community training in water-management-plan writing, and the return of consolidated data and plans to the LAOs.

3.4 Local Capacity-Development Framework

Capacity-building targeted six skill areas: water source mapping, construction of a water-use calendar, water balance calculation, use of geolocation technology, report writing, and project-proposal writing. Training followed a four-step process – hybrid learning (combining onsite instruction with on-demand video resources), field data collection, digital data entry, and project-plan development – organized around four capacity dimensions. The political dimension focused on strengthening WUOs' and communities' standing to engage LAOs and higher-level water authorities. The economic dimension focused on skills for identifying and costing feasible water-management interventions. The social dimension focused on strengthening collective organization and consensus-building within and across WUOs. The digital-technology dimension focused on platform use, GIS-based data collection, and digital reporting. Implementation followed a coaching-and-monitoring model whose first step was a gap analysis comparing each community's current capacity against the capacity required to independently produce and submit water-management plans, with subsequent coaching sessions targeting the identified gaps.

3.5 Digital Platform Development

The Waterplan RRU platform (<https://waterplanrru.org>) was developed following a systems development life cycle (SDLC) approach with three phases. The analysis phase produced use case, activity, and sequence diagrams for core functions – user login, knowledge creation, testing and evaluation, and data management – which were consolidated into a software requirements specification (SRS). The design phase specified the system architecture (front-end interface, application layer, database, GIS services, and API layer), user-interface design, and database design. The testing phase covered software functionality testing, data accuracy testing, data integrity testing, and security testing, culminating in the nine-expert architecture evaluation described in Section 4.5.

3.6 Instruments

Data collection used a combination of GIS-based field survey tools; an unstructured in-depth interview guide for initial stakeholder consultation; a structured interview guide covering four question areas, administered to representatives of the six participating WUOs; a focus-group discussion guide used with community leaders; and a workshop discussion guide incorporating after-action review (AAR) to capture lessons learned at the close of major project activities.

3.7 Data Analysis

Quantitative survey data were analyzed using arithmetic means ($\bar{X} = \Sigma X/N$) and standard deviations, interpreted against a five-point Likert scale (4.50–5.00 highest, 3.50–4.49 high, 2.50–3.49 moderate, 1.50–2.49 low, 1.00–1.49 lowest). Expert-panel agreement on the platform architecture was interpreted using a separate five-point agreement scale (4.51–5.00 highest agreement, 3.51–4.50 high agreement, 2.51–3.50 moderate agreement, 1.51–2.50 low agreement, 1.00–1.50 lowest agreement). The index of item-objective congruence ($IOC = \Sigma R/N$) was used to validate instruments prior to field use.

4. Results

4.1 Context Evaluation

Across the four subdistricts, water quality was rated “fair” by community respondents. Six recurring problem categories were reported consistently across the six participating WUOs: dry-season water shortage (typically February–April), rainy-season flooding, saltwater intrusion from the Bang Pakong River, shallow and silted canals, damaged pipe systems (primarily aging PVC pipe), and obstructions in waterways, compounded by a general lack of spatial data on local water sources. Underlying contextual pressures included rising agricultural and industrial water demand, increasing salinity linked to Irrigation Department salinity-control operations on the Bang Pakong River, water-use conflicts among users, land-tenure constraints limiting new reservoir construction, and shortages of government staffing and equipment for local water-infrastructure maintenance.

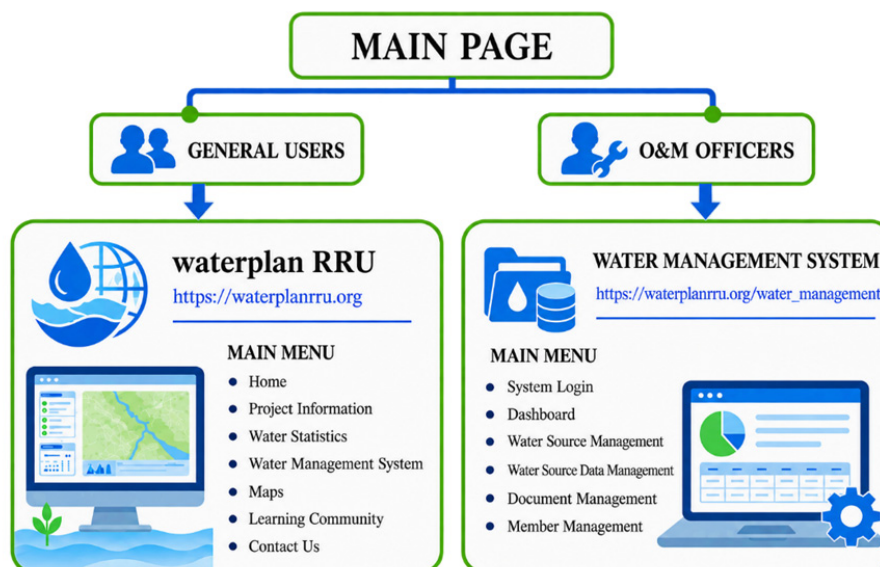


Figure 2. Platform is a web-based decision-support system for Waterplan RRU

The Figure 2 platform is a web-based decision-support system for water resource management that provides role-based access for General Users and O&M Officers. General Users can access water resource information, statistics, maps, projects, and learning resources through the Waterplan RRU platform, while O&M Officers can manage water sources, datasets, documents, members, and system information through the Water Management System. This structure supports systematic information management and provides reliable data for planning, monitoring, and evidence-based decision-making in water resource management.

4.2 Area Water Potential, Demand, and Balance

Field survey identified 102 water sources across the four subdistricts, with a combined water potential of 7,318,430 cubic meters as detailed and shown in Table 1. An earlier-stage estimate reported in the project's executive summary, 101 sources and 7,313,656 m³, was superseded by this more detailed subdistrict-level inventory, which is used as the authoritative figure throughout this paper.

Table 1. Water source potential by subdistrict

Subdistrict	Number of Sources	Water Potential (m ³)
Samet Tai	36	3,114,501
Samet Nuea	26	1,199,032
Khleng Khut	21	1,159,798
Plaeng Yao	19	1,845,099
Total	102	7,318,430

The total water demand across the four subdistricts, encompassing both domestic and agricultural uses, was estimated at 83,241,498.95 m³. Agricultural use constituted the predominant share of total water demand in all four subdistricts. Although industrial water use was identified as a contextual factor contributing to overall water demand, no separate quantitative data on industrial water consumption were reported in the study. The detailed water demand estimates are presented in Table 2.

Table 2. Water demand by subdistrict

Subdistrict	Household (m ³)	Agricultural (m ³)	Total (m ³)
Samet Tai	97,528.00	24,159,753.05	24,257,281.05
Samet Nuea	91,250.00	10,301,622.08	10,392,872.08
Khleng Khut	41,555.25	11,169,169.56	11,210,724.81
Plaeng Yao	180,529.00	37,200,092.01	37,380,621.01
Total	410,862.25	82,830,636.70	83,241,498.95

Comparing demand against potential yielded a combined water balance deficit of 75,923,068.95 m³ (Figure 3). Every subdistrict showed demand substantially exceeding locally available water potential, with Plaeng Yao showing the largest absolute deficit.

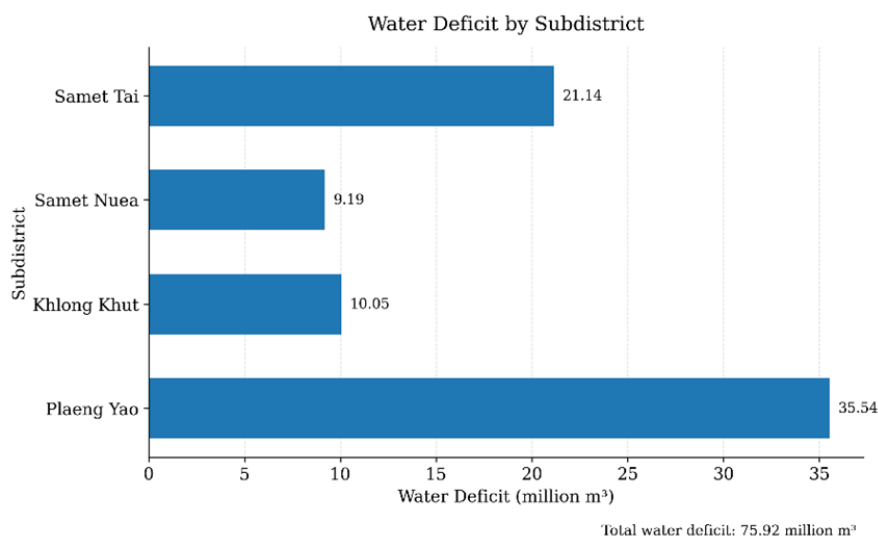


Figure 3. Water deficit by 4 subdistrict in Chachoengsao Province, Thailand

4.3 Results of the Assessment of Assigned Training Tasks Using the Digital Platform for Planning and Decision-Making in Water Resource Management

The researchers assessed the assigned tasks completed by trainees who were trained through the digital water management platform system. Following the training, scores were assigned using a standardized five-level rating scale based on Likert's Scale, with interpretation criteria shown in Table 3.

Table 3. Results of the water balance analysis competency assessment

Evaluation Item	Mean ($\bar{x}$)	S.D.	Level
Competency 1: Water Balance Analysis Competency			
1. Skill in identifying and locating water sources	4.49	0.40	High
2. Skill in mapping water-flow diagrams	4.44	0.45	High
3. Skill in calculating water source volume	4.31	0.46	High
4. Skill in calculating water usage volume according to land-use pattern	4.39	0.47	High
5. Skill in analyzing water balance	4.59	0.45	Very High
Total for Competency 1: Water Balance Analysis	4.44	0.45	High

The knowledge and understanding required of the trainees, together with the practical skills and attitudes necessary to achieve the intended learning outcomes, were identified as key components of the water balance analysis competency. The results of the competency analysis are presented in Table 4.

Table 4. Learning outcomes for the water balance analysis competency

Competency	Know & understand	Skills	Attitude
1. Water balance analysis	- Meaning and types of water sources - Principles of water-quantity and water-use calculation - The concept of water balance	- Identify and locate water sources - Draw water-flow diagrams - Calculate water-source volume and water use - Analyze water balance	- Awareness of the importance of water sources in the community - Sense of responsibility for water resource management
Learning Outcome	Task to Perform	Evaluation Criteria	Learning Activity
1.1 Correctly identify water sources and their locations	- Draw a map identifying water sources with coordinates	- Accuracy of location / completeness of water sources	- Analyze actual community maps and practice drafting water-source diagrams
1.2 Draw water-flow diagrams consistent with actual conditions	- Construct a diagram of water flow from the source to the target area	- Logical consistency of the water route / clarity of the diagram	- Use Google Maps/topographic maps to assist in drawing the diagram
Learning Outcome	Task to Perform	Evaluation Criteria	Learning Activity
1.3 Calculate water-source volume	- Apply calculation formulas based on the dimensions of a pond or reservoir	- Accuracy of calculation / actual measurement	- Measure an actual pond in the field, then use the data to calculate volume (width x length x depth)
1.4 Calculate water-use volume based on land-use area	- Analyze water use in agricultural/community areas	- Accuracy of estimation / reference to supporting data	- Analyze real-world cases, such as community ponds or canals
1.5 Analyze water balance for the area	- Compare water-source volume against water-use volume	- Correct analysis, consistent with actual data	- Practice using the water-balance worksheet, then analyze and discuss the results

The knowledge and understanding required of the trainees, together with the practical skills and attitudes necessary to achieve the intended learning outcomes, were identified as key components of competency in using the digital water-management platform. The results of the competency analysis are presented in Tables 5 and 6.

Table 5. Results of the digital water management platform use competency assessment

Evaluation Item	Mean ($\bar{x}$)	S.D.	Level
Competency 2: Digital Water Management Platform Use Competency			
2.1 System Use Skills			
1. Skill in registering and logging in to the digital water-management platform	4.48	0.41	High
2. Skill in entering water-storage data (reservoirs, ponds)	4.34	0.45	High
3. Skill in entering water-source data	4.30	0.46	High
4. Skill in entering width, length, depth, and location data	4.39	0.47	High
5. Skill in photographing water sources and uploading images to the system	4.25	0.45	High
6. Skill in entering land-use and topographic data	4.13	0.38	High
7. Skill in entering basic GIS data for water management	4.47	0.41	High
2.2 Lifelong Learning Skills			
8. Skill in problem-solving in data management	4.30	0.45	High
9. Applying acquired knowledge to writing water-management project proposals	4.32	0.46	High
10. Selecting and using technology to solve problems creatively and ethically	4.29	0.46	High
11. Ability to transfer knowledge to others in using the digital platform	4.31	0.44	High
Total for Competency 2: Digital Water Management Platform Use	4.33	0.44	High

Table 6. Learning outcomes for the water management plan writing competency

Competency	Know & understand	Skills	Attitude
1. Water management plan writing	- Principles of project planning and design - Steps for analyzing the water situation - Budget management and community participation	- Analyze water problems in the area - Draw maps and management diagrams - Write work plans and manage budgets - Promote participation and teamwork	- Respect for others' opinions - Commitment to working for the collective good - Willingness to listen and adjust plans as circumstances change
Learning Outcome	Task to Perform	Evaluation Criteria	Learning Activity
1.1 Analyze subdistrict-level water problems	- Write a report analyzing the water situation based on community data	- Use of supporting data, with proposed solutions	- Interview community leaders and analyze real data
1.2 Draft a community water-management diagram	- Produce a hand-drawn concept diagram of the proposed water-management approach	- Diagram clear, complete, and appropriate to the context	- Work in groups to draft plans together with the community
1.3 Draw a map of the community's water flow path	- Prepare a map showing the flow of water within the area	- Accuracy, consistent with actual topography	- Conduct an actual field survey / review aerial photographs
1.4 Design structural interventions such as weirs or reservoirs	- Draft a basic plan for a small-scale project	- Approach clear and feasible	- Use a design process to develop proposed solutions
1.5 Plan the budget for a water project	- Prepare a budget plan to accompany the project	- Items clear, carefully considered, and appropriate to the context	- Analyze an actual project budget, then simulate drafting a new plan

Across the two competency domains, a paired-samples t-test was conducted to compare trainees' knowledge scores before and after the training. The results indicated a statistically significant improvement in knowledge, with the mean score increasing from 8.53 (SD = 3.08) on the pre-test to 10.50 (SD = 1.58) on the post-test, based on a maximum possible score of 15, $t(64) = 7.21, p < .01, n = 65$.

4.4 Platform Design

The system analysis resulted in use-case specifications covering four core functional areas: user authentication and access management, knowledge development through community-based water data collection and mapping, testing and evaluation, and data management. These specifications informed the system architecture presented in Section 3.5. The resulting platform, WaterplanRRU, was developed as a mobile-first web application and comprises a front-end interface, application layer, relational database, GIS services for spatial data management, and an API layer that facilitates data exchange among community-collected data, Local Administrative Organization (LAO) records, and, prospectively, the national TWP system (Figure 4).

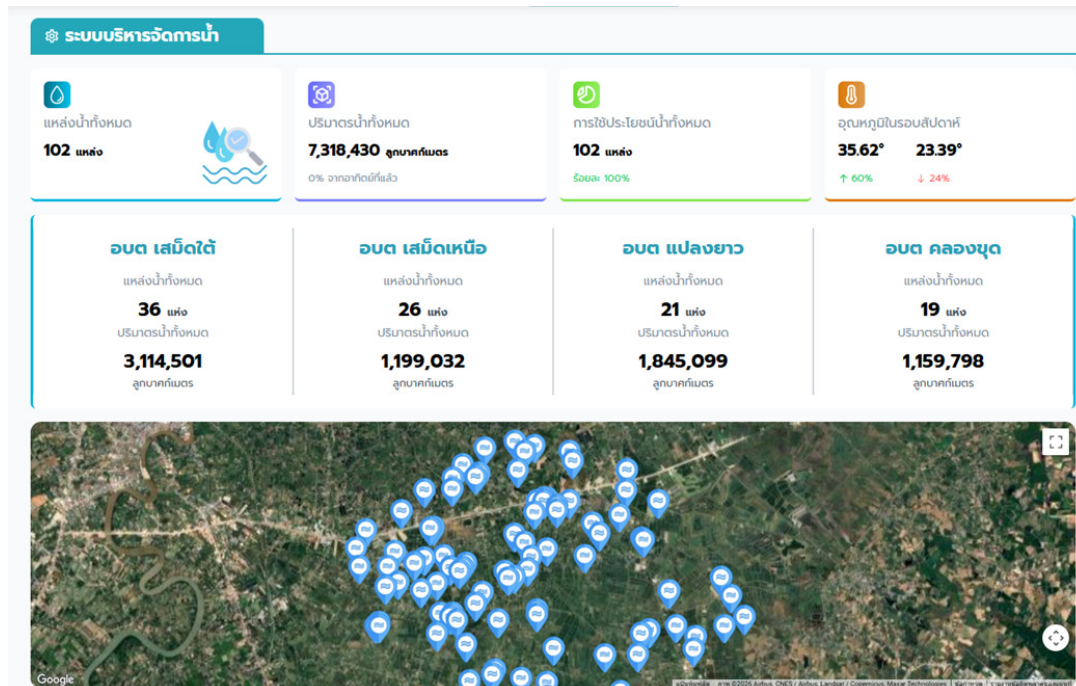


Figure 4. System web application example screen WaterplanRRU

4.5 Architecture Evaluation

The Nine domain experts evaluated the platform architecture across three functional domains: the ecosystem framework and system scope, the user management system, and the knowledge and learning management system, comprising a total of 15 evaluation items. The mean scores for individual items ranged from 4.37 to 4.80. Fourteen items were rated at the “highest” level according to the expert-agreement scale, while the system scope and coverage item received a “high” rating, with a mean score of 4.37. The overall mean score for the platform architecture was 4.61 (SD = 0.48), indicating the “highest” level of suitability according to the established interpretation criteria.

4.6 Process Change and Project Outcomes

Platform adoption was associated with a shift in how water-management projects moved from community identification to LAO planning documents (Table 7).

Table 7. Process change following platform implementation

Process		Before Platform	After Platform
Community/group consultation		LAO-convened, village-level meetings	WUOs/communities self-organize and propose plans directly to the LAO
Project detail documentation		Recorded and compiled solely by LAO staff	Co-written by community and LAO staff using the shared platform database
LAO plan preparation		Built from consultation-meeting notes	Drawn directly from the digital-platform database, aligned with community needs

Regarding TWP submission specifically, the four LAOs fell into two groups at project outset. Samet Tai Subdistrict AO already held TWP system credentials and had submitted a project for fiscal-year 2027 budget consideration,

though it had not yet been funded, for three identified reasons: staff turnover affecting continuity of login-credential holders, unclear alignment between the proposed project and TWP budget categories, and incomplete supporting documentation. The remaining three LAOs (Samet Nuea, Plaeng Yao, and Khlong Khut) had not been able to make effective use of the TWP system – Plaeng Yao held credentials but had never used them, while Khlong Khut and Samet Nuea had lost or forgotten their credentials, again due to staff turnover. Following hands-on training delivered by NWRO Region 2, all four LAOs successfully recorded water-management projects in the TWP system for fiscal year 2028.

4.7 Competency and Satisfaction Outcomes

The user satisfaction was assessed among 94 respondents from the four participating Local Administrative Organization (LAO) areas, comprising 63 male and 33 female respondents. The results indicated that all five dimensions were rated at a high level, with mean scores ranging from 4.15 to 4.25. Security received the highest mean score ($M = 4.25$, $SD = 0.58$), followed by usability ($M = 4.24$, $SD = 0.61$) and performance ($M = 4.22$, $SD = 0.61$). Functional requirements and functional testing received the same mean score ($M = 4.15$), with standard deviations of 0.57 and 0.53, respectively. These findings indicate that the developed platform achieved a high level of user satisfaction across functional requirements, usability, performance, security, and functional testing dimensions (Figure 5).

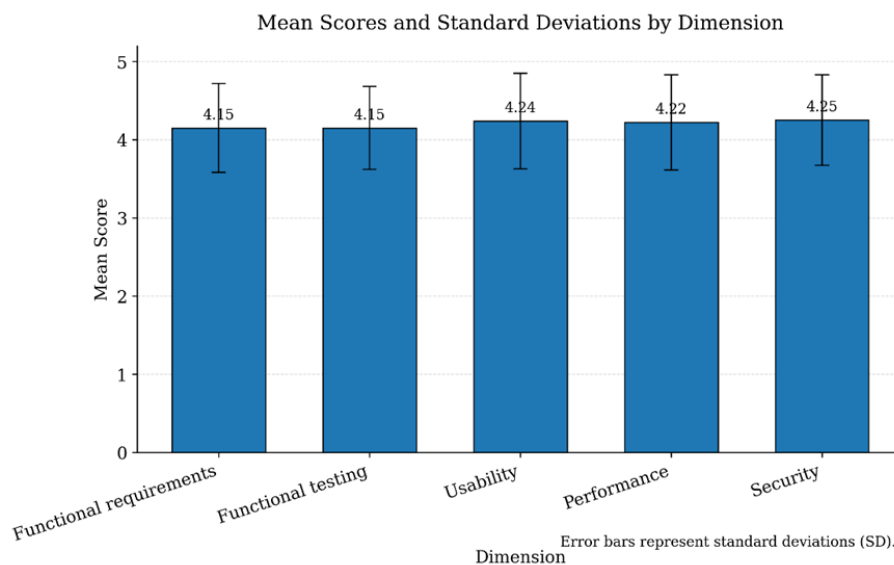


Figure 5. Mean scores and standard deviations by dimension

4.8 Knowledge-Return Process

Data and findings were returned to communities and LAOs through a five-step process: field survey and data collection across the four LAO areas; compilation of collected data into the platform's information system, organized into categories covering current infrastructure, urgent needs, and short-, medium-, and long-term project candidates; joint verification workshops with researchers, WUOs, and community members to confirm accuracy, fill data gaps, and refine project designs; formal delivery of consolidated outputs to the LAOs, including water-source maps and project lists with stated objectives and expected impacts, for direct use in local development planning; and a concluding lessons-learned exercise identifying follow-up priorities, including formal WUO establishment where not yet in place, longer-term area-development projects, a subdistrict-level water database, and an ongoing coordination network linking NWRO, LAOs, and communities.

5. Discussion

The Four key themes emerged from the findings. First, the water balance analysis identified a combined water deficit of 75,923,068.95 m³ against a locally available water potential of 7,318,430 m³, providing an important empirical basis for integrated water resources management (IWRM) at the subdistrict level. The main contribution of the study in this regard was the systematic collection and organization of quantitative baseline data, which supports evidence-based

decision-making in local water-resource management. Second, the high expert evaluation score for the platform architecture ($M = 4.61$) and the significant improvement in trainee competency ($t = 7.21$, $p < .01$) demonstrate that a locally adapted digital platform combined with structured training can support standardized community-based water-data collection. This finding is consistent with digital government and data governance research, which emphasizes that effective digital systems require both appropriate technological infrastructure and the capacity of users to apply the system in practice. Third, the transition from LAO-compiled consultation records to platform-based documentation developed jointly by communities and LAOs, followed by the submission of four approved TWP projects, demonstrates a clear connection between community-generated water data and formal planning and funding processes. This outcome addresses the governance gap identified in Section 1.1, where a formal mechanism for community participation in water-resource planning existed but had been rarely used in practice. Fourth, the five-step knowledge-return process supported social learning by returning verified water-resource data and planning information to communities and LAOs, thereby strengthening shared understanding, trust, and ownership among stakeholders. These findings also indicate that the long-term sustainability of the platform depends on institutional capacity in addition to technological infrastructure. The TWP submission history in Samet Tai illustrates this issue, as the LAO had system access and had previously submitted a project but remained unfunded following staff turnover and uncertainty regarding project-category alignment. This experience highlights the importance of maintaining trained personnel and institutional knowledge to ensure continued use of the platform. Based on these findings, the study proposes a four-dimensional sustainability framework comprising technology, data, personnel, and network. The technology dimension involves Rajabhat Rajanagarindra University hosting and maintaining the platform as the infrastructure provider, while the data dimension emphasizes continued ownership of water-resource data by LAOs. The personnel dimension focuses on developing community-level and LAO-level innovators who can maintain platform use and support local water-resource management beyond the project period. The network dimension emphasizes continued coordination among the National Water Resources Office (NWRO), LAOs, and communities to support sustained data sharing, planning, and collaborative water-resource management.

6. Conclusion

This study assessed water resource potential and demand, built local capacity, and developed and evaluated a digital platform for water resource planning across four subdistricts of Chachoengsao Province. Seven principal findings summarize the results. The four subdistricts hold a combined water potential of 7,318,430 m³ across 102 identified sources, with water quality generally rated “fair.” Total water demand across the four subdistricts is 83,241,498.95 m³, driven predominantly by agricultural use. The resulting water balance deficit is 75,923,068.95 m³, with Plaeng Yao subdistrict showing the largest shortfall. A mobile-first digital platform, Waterplan RRU, was successfully developed and rated at the highest suitability level by expert evaluators. All six participating WUOs produced water-management plans that were submitted to their LAOs for inclusion in 2023–2027 local development plans, and all four LAOs subsequently recorded water-management projects in the national TWP system. Local and LAO-level innovators – including LAO executives, Civil Engineering staff, and policy-and-plan-analysis staff capable of using the platform’s GIS dashboard – were identified and developed over the course of the project. Community-level innovators, including WUO members and community leaders capable of continuously updating platform data and translating it into water-management plans, were likewise identified and positioned to sustain the platform’s use going forward.

For future research, five directions are recommended: strengthening decision-support and forecasting functions, including GIS-based analysis and real-time reporting; expanding data accessibility through multiple formats (text, image, video, infographic, audio) and AI-assisted search and summarization within a simplified interface; developing lifelong-learning modules tailored to different user types, incorporating gamification; integrating the platform with external systems such as the Royal Irrigation Department and Meteorological Department and the NWRO, with module customization by area type and a public incident-reporting channel; and developing the platform toward an open knowledge repository. For policy, three recommendations follow directly from the findings: establishing a real-time, IoT-linked water-data center paired with a formal decision-support system; strengthening citizen-participation mechanisms through app-based issue reporting and social-media-based public sentiment analysis; and sustaining capacity-building through incentive mechanisms, including performance-linked budget allocation, innovation awards, and incorporation of platform-related indicators into staff performance metrics. The ethical approval number for this research is HREC 020-08-2026, issued by the Faculty of Management, Shinnawatra University.

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Author Contributions

All authors contributed equally to the conception, design, data collection, analysis, and writing of this study, and approved the final manuscript.

Consent for Publication

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Declaration of Competing Interest

The authors declare that they have no competing interests.

Informed Consent

Informed consent was obtained from all individual participants included in the study, including community leaders, water user organization representatives, and local administrative organization officers.

Ethics Approval

This study was conducted in accordance with the research ethics requirements of the funding agency.

Data Availability Statement

The Waterplan RRU platform and its underlying database are maintained by Rajabhat Rajanagarindra University. Raw survey data supporting the findings of this study are available from the corresponding author upon reasonable request.

Use of Artificial Intelligence

Not applicable.

Supplementary Materials

No additional data are available.

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